Deep Hybrid Model to classify Satellite Image using Vegetation Indices feature set

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2024 1st International Conference on Cognitive, Green and Ubiquitous Computing (IC-CGU) | 979-8-3503-6178-0/24/$31.00 ©2024 IEEE | DOI: 10.1109/IC-CGU58078.2024.10530816

***Abstract*— A variety of information about the region of interest can be found in satellite images through multispectral and panchromatic elements. For several purposes, including mapping of vegetation and crop, accurate classification of satellite imagery is necessary. Hybrid classifiers, providing techniques to combine and integrate many classifiers, have proven significant efficiency in classification of satellite images in recent years. In the way, this paper attempts to propose a satellite image classification model. The satellite image is first pre-processed by turning it into a grayscale image, applying a histogram to sharpen the image's edges, and eliminating any noise. Then the suitable vegetation index feature set is extracted from the preprocessed image, which includes EVI, NDVI, WDRVI and ENDVI. Finally, a hybrid model is proposed to classify the image, which is the combination of Improved CNN and LSTM that trained with the feature set. Lastly, the suggested work's efficiency is verified with regard to of sensitivity, accuracy, and specificity compared to the traditional methods.**

Keywords—Satellite Image, Vegetation Index, VI, NDVI, Hybrid Model

I. INTRODUCTION

In recent years, High spectrum image processing, satellite imaging, and surveillance by air are three fields where technology has developed most significantly and rapidly [1]. For several earth observation activities, including the evaluation of warming temperatures, the analysis of desertification and deforestation, and the identification of changes in land across aims on producing specific maps of the Earth's surface [2]. VI (vegetation index) are frequently used as data processing system to distinguish among vegetated as well as non-vegetated items on the earth surface. When employing a satellite recognizing technique to evaluate vegetation covering, the vegetative Vigor test, and plant development factors, VIs is a simple and effective approach. VARI (Visible atmospheric resistant index), GLI (green leaf index), and VI green are three examples of VI methods. For analysis, these methods just need RGB channel data. The majority of relevant research has demonstrated that these VI approaches produce accurate results, particularly for data from agricultural areas [3]. However, in urban locations with a variety of object classifications, green non-vegetation objects are subject to error in decision-making, which results in incorrect classification of objects [4]. To increase the sensitivity of recognizing green vegetation, RGB and NIR band methods are combined. Further investigation and attention are needed to address the limitations of the vegetation identification method for remote sensing utilizing RGB images. The major contribution of this work is to address the restriction of current satellite image classification with VI, a hybrid model is proposed with the inclusion of models like Improved CNN (Convolutional neural network) and LSTM (Long short-term memory network) based on the feature set like EVI (enhanced vegetation index), NDVI (Normalized difference vegetation index), WDRVI (wide dynamic range vegetation index), ENDVI (enhanced normalized difference vegetation index) [5].

II. PROPOSED FRAMEWORK FOR CLASSIFICATION

In this paper, a hybrid deep learning model for satellite image classification is proposed. with vegetation Indices based on the suitable feature set. The step-by-step procedure followed is (a) Preprocess of the satellite image (b) Extraction of Relevant feature Set (c) Classification: Hybrid model combining Improved CNN and LSTM. The architecture of the hybrid model is shown in Fig 1.

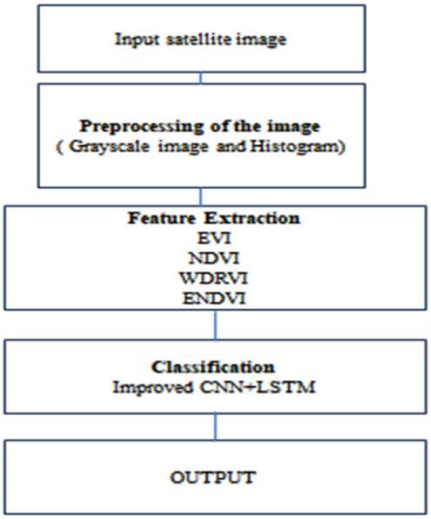


Fig 1: Architecture of the hybrid model

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Preprocessing improves the image's edges as well as smoothing out and removing any noise that existed in the original image. Initially, the input image is converted into the gray scale image in order to decrease the color complexity [6] and used to apply histogram in order to get a lot of gray-level information and a broad dynamic range. The spectral indices used to extract features in this paper are EVI, NDVI, WDRVI and ENDVI to highlight area of vegetation in the preprocessed image. Classification process is based on the feature attributes F employed to represent the data image classes. As an outcome, this work suggests a hybrid model that combines the LSTM and Improved CNN models. The feature F is simultaneously fed as the input to the Improved CNN as well as to LSTM. in this classification process. The ultimate classification output is then assessed using the mean of the intermediate results from both the LSTM and the Improved CNN. An additional Convolution block 3 is created in Improved CNN as shown in Fig 2.

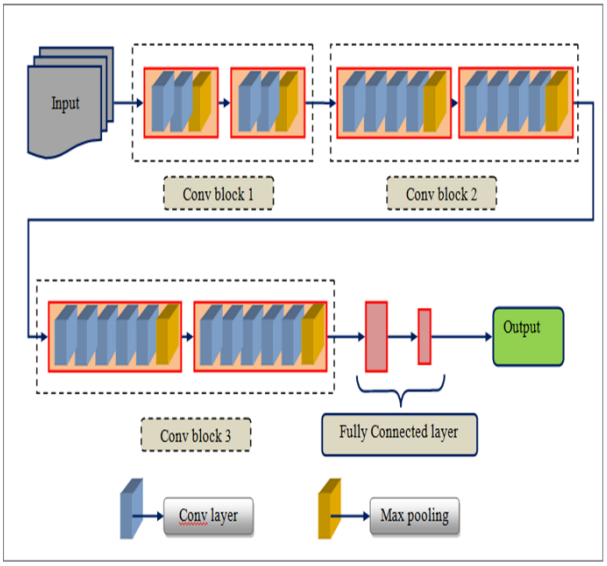


Fig 2: Improved CNN structure

The configuration of LSTM and RNN are similar, but the LSTM has memory cell instead of RNN hidden layer. LSTM can solve the gradient disappearance problem based on the long-time data storing memory cell. The memory cell of LSTM has single unit and three gates such as input gate, output gate and hidden gate. These three gates are the information gate. It controls the transmission information of neurons and also the transmission information among the existing neuron and the subsequent neuron. It has activation function to controls them by multiplication factor [7]. Structure of LSTM is shown in Fig. 3. In this work, the final extracted feature F is subjected as the input feature to the input layer of LSTM, which is represented as $x_t$at time moment t, h_{t-1} denotes the hidden layer, c_{t-1}indicates the memory cell of LSTM, $g$represents the input function, $h$represents the output function,$f$denotes the activation function. Equation (1) to (6) illustrates the calculation of LSTM intermediate output. Initially, at time moment t, three gates are calculated. Where, I_t, O_t andf_t are the input, output and forget gates respectively at time t.

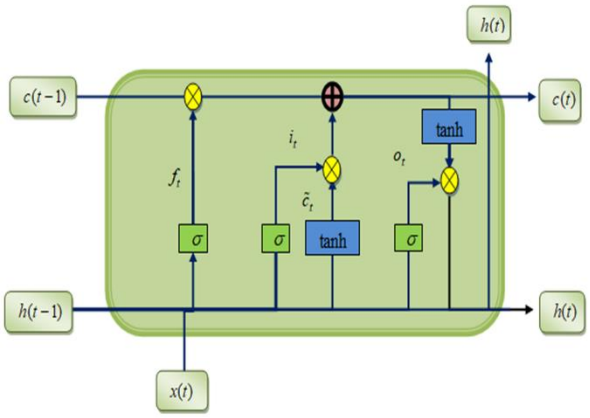


Fig 3: Overall Framework of LSTM

P_I, P_O, $P_f$are weights of input, output and forget gate respectively.B_I, B_O, B_f are the bias vector of the respective gates. Then, the memory cell $c_t$is computed in Equation (4) and (5). Here, $\bar{c}_t$denotes the memory cell without forget gate. The hidden layer is calculated in Equation (6). From the calculation, LSTM can store the most significant data in the memory cell and also it solves the gradient disappearance problem.

$$I_t = \sigma(P_{Ix}x_t + P_{Ih}h_{t-1} + B_I) \quad (1)$$
$$O_t = \sigma(P_{Ox}x_t + P_{Oh}h_{t-1} + B_O) \quad (2)$$
$$f_t = \sigma\left(P_{fx}x_t + P_{fh}h_{t-1} + B_f\right) \quad (3)$$
$$\bar{c}_t = tanh(P_{cx}x_t + P_{ch}h_{t-1} + B_c) \quad (4)$$
$$c_t = f_t \times c_{t-1} + I_t \times \bar{c}_t \quad (5)$$
$$h_t = O_t \times tanh(c_t) \quad (6)$$

Finally, both the intermediate output obtained from the deep hybrid model (Improved CNN as well as LSTM) are combined to get mean function in order to evaluate the classified output.

III. RESULT ANALYSIS ON HYBRID MODEL BASED IMAGE CLASSIFICATION

The proposed LANDSAT 8 image was classified using the Earthexplorer dataset [8] and classification was implemented in "PYTHON 3.7".

A. Performance Analysis

The Improved CNN+LSTM and the traditional algorithms were analyzed in terms of precision, NPV, specificity, FPR, accuracy, F-measure, FNR, MCC and sensitivity. The Improved CNN+LSTM was compared with the traditional classifiers like SVM (support vector machine), Bi-GRU (bidirectional gated recurrent unit), RF (random forest), RNN (recurrent neural network) and DT (decision tree).

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B. *Assessment of positive metric*

The positive metric examination on Improved CNN+LSTM over Bi-GRU, SVM, DT, RF and RNN for satellite image classification is exposed in Fig 4. The model should receive higher positive metric ratings for the precise classification of satellite images. In particular, the accuracy of Improved CNN+LSTM approach is 0.958 at the training rate 80, mean while the Bi-GRU, SVM, DT, RF and RNN recorded lower accuracy metric ratings. Similarly, the highest precision is yielded utilizing Improved CNN+LSTM methodology than the other conventional strategies. Furthermore, the sensitivity of Improved CNN+LSTM methodology is 0.947 (Learning Percentage=90), even though the traditional methods acquired lesser sensitivity values, notably, Bi-GRU=0.885, SVM=0.871, DT=0.853, RF=0.796 and RNN=0.804, correspondingly. Likewise, the specificity of the Improved CNN+LSTM method is extremely higher in all the training rates. This assessment verified the enhanced effectiveness of Improved CNN+LSTM methodology for classifying the satellite images. This advancement is owing to the novel classifier strategy for classifying the satellite images in an optimal manner.

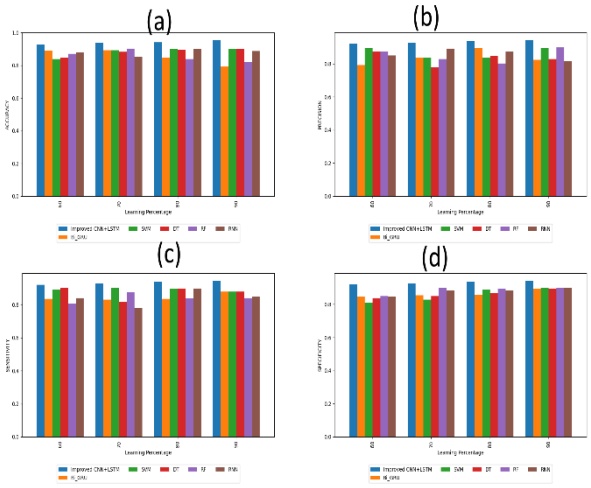


Fig 4: Examination on Improved CNN+LSTM and conventional strategies regarding positive metric for satellite image classification. (a) Accuracy (b) Precision (c) sensitivity (d) specificity

C. *Assessment of Negative metric*

The evaluation on Improved CNN+LSTM over Bi-GRU, SVM, DT, RF and RNN regarding negative metric for satellite image classification is described in Fig 5. Further, the negative metric value should be higher for the accurate classification of satellite images. Similarly, the Improved CNN+LSTM recorded minimized negative metric ratings over the conventional strategies. Mainly, the FNR (false negative rate) of the Improved CNN+LSTM scheme is 0.076 at the training rate 90%, whilst the Bi-GRU is 0.135, SVM is 0.123, DT is 0.147, RF is 0.118 and RNN is 0.149, correspondingly. In addition, the FPR of the Improved CNN+LSTM methodology is minimal in the entire training rates, though the Bi-GRU, SVM, DT, RF and RNN obtained higher FPR (false positive rate) values. Thus, the Improved CNN+LSTM approach has indicated its advancement by acquiring minimized negative metric values. This has paved the way for the accurate classification of satellite images.

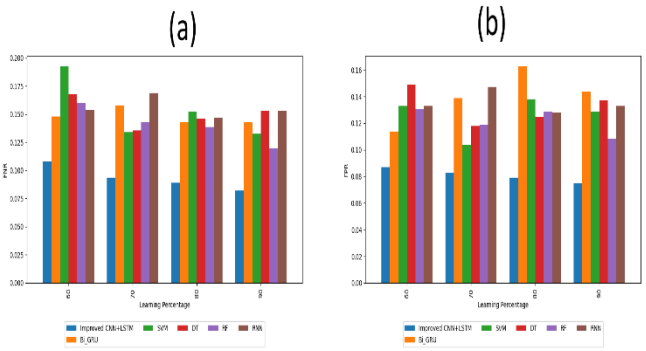


Fig 5: Examination on Improved CNN+LSTM and conventional strategies regarding negative metric for satellite image classification (a) FNR (B) FPR

D. *Assessment on other metric*

The other metric estimation on Improved CNN+LSTM is contradicted over Bi-GRU, SVM, DT, RF and RNN for satellite image classification is explained in Fig 6. The other metric values should be higher for effective classification of satellite images. Mainly, the greatest F-measure is acquired utilizing Improved CNN+LSTM method is 0.954, though the traditional schemes offered lower F-measure values, such as, Bi-GRU, SVM, DT, RF and RNN. Furthermore, the MCC (Mathhew's correlation coefficient) and NPV (negative predictive value) of the Improved CNN+LSTM methodology is 0.927 and 0.936 at the training rate 80, whereas the Bi-GRU, SVM, DT, RF and RNN hold minimal values. Therefore, the Improved CNN+LSTM model findings on other metric indicates that it obtained higher ratings in the entire training rates where this is attained by introducing the novel classifier model during the classification of satellite images.

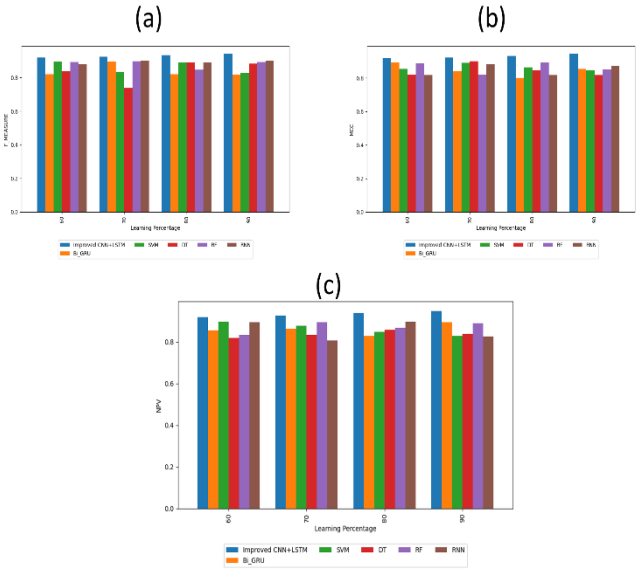


Fig 6: Examination on Improved CNN+LSTM and conventional strategies regarding other metric for satellite image classification (a)F-measure (b) MCC (c) NPV

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E. Ablation analysis on Improved CNN and LSTM

Ablation evaluation on Improved CNN+LSTM, model without improved CNN and model without feature extraction for satellite image classification is illustrated in Table I. Here, train the Improved CNN+LSTM approach repeatedly while maintaining the feature extraction and improved CNN. Similarly, assess the impact of Improved CNN+LSTM methodology. In general, accomplish the experiments while keeping the feature extraction and improved classifier approach to the Improved CNN+LSTM technique. Moreover, the precision of the Improved CNN+LSTM scheme is 0.928, model without improved CNN is 0.848 and model without feature extraction is 0.830. Additionally, the FPR of the Improved CNN+LSTM=0.083, model without improved CNN=0.105 and model without feature extraction=0.129.

TABLE I. ABLATION STUDY ON IMPROVED CNN+LSTM, MODEL WITHOUT IMPROVED CNN AND MODEL WITHOUT FEATURE EXTRACTION FOR SATELLITE IMAGE CLASSIFICATION

Methods	**Model without improved CNN**	**Model without feature extraction**	**Improved CNN+LSTM**
Sensitivity	0.848	0.903	0.927
FPR	0.105	0.129	0.083
Specificity	0.818	0.864	0.924
F measure	0.857	0.905	0.924
FNR	0.182	0.136	0.094
Accuracy	0.894	0.874	0.936
MCC	0.783	0.856	0.922
Precision	0.848	0.830	0.928
NPV	0.895	0.847	0.926

F. Statistical Evaluation on Accuracys

The statistical assessment on Improved CNN+LSTM over Bi-GRU, SVM, DT, RF and RNN for satellite image classification is exposed in Table II. The metaheuristic proceedings are undependable; For this, five distinctive types of statistical metrics—including mean, minimum, standard deviation, median, and median—are examined[9]. Further, the Improved CNN+LSTM acquired the accuracy of 0.928 at the minimum statistical metric, whilst the Bi-GRU is 0.793, SVM is 0.839, DT is 0.850, RF is 0.819 and RNN is 0.853, correspondingly.

TABLE II. STATISTICAL ASSESSMENT ON IMPROVED CNN+LSTM AND TRADITIONAL STRATEGIES WITH REGARD TO ACCURACY

Statistical Metrics	Improved CNN+LSTM	Bi-GRU	SVM	DT	RF	RNN
Mean	0.940	0.856	0.883	0.882	0.857	0.880
Minimum	0.928	0.793	0.839	0.850	0.819	0.853
Median	0.940	0.869	0.897	0.889	0.854	0.884
Maximum	0.953	0.893	0.901	0.900	0.903	0.899
Standard Deviation	0.009	0.041	0.026	0.020	0.032	0.017

IV. CONCLUSION

This paper proposed a deep hybrid model for satellite image classification with vegetation Indices based feature set. At first, the satellite image is pre-processed enhance the edges of the image and eliminate noise in the image. Then the suitable feature set was extracted from the pre-processed image. EVI, NDVI, WDRVI and ENDVI are the features extracted from the pre-processed image. At last, a hybrid model that combined an improved CNN along with an LSTM model was proposed in order to improve the accuracy of classification using the final extracted feature set. Furthermore, the accuracy, specificity and sensitivity of the performance classifier were measured, and the hybrid model's experimental findings were compared to those of other classifiers like LSTM, Bi-GRU, RNN, SVM, RNN, and ANN in order to achieve higher accuracy. This research's main contribution will be to solve the limitations of the present Vegetation Index classification of satellite images and to offer hints that can aid in real-time crop and vegetation management.

ACKNOWLEDGMENT

I am grateful to Centurion University of Technology and Management for offering a variety of research facilities for this project.

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